Hedging the AI Singularity

Andrew Y. Chen

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Abstract

We develop a model where AI equity commands high valuations because it hedges against a negative AI singularity—an explosion of AI development that devastates the representative investor through labor displacement. When transformative AI causes consumption to collapse while AI firms capture economic surplus, publicly traded AI assets provide partial insurance through negative covariance with marginal utility. This hedging mechanism can justify elevated valuations without requiring beliefs about extraordinary dividend growth. The analysis suggests financial market solutions—such as broadening AI equity ownership—deserve more attention in policy discussions about AI catastrophe risk. Unlike previous work, this short paper is written by prompting LLMs.

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1 Introduction

Artificial intelligence is advancing at an unprecedented pace. In late 2024, OpenAI released o1, a reasoning model that dramatically improved performance on complex mathematical and coding tasks. Within months, DeepSeek's R1 matched o1's capabilities using substantially fewer computational resources (DeepSeek-AI et al., 2025). By early 2025, OpenAI's o3 achieved 87.5% accuracy on the ARC-AGI Challenge, a benchmark designed to measure general intelligence through novel problem-solving (Chollet et al., 2024). Meanwhile, Waymo's autonomous vehicles have logged tens of millions of miles on public roads, demonstrating how AI systems are transitioning from laboratory curiosities to deployed technologies reshaping transportation, logistics, and urban planning.

These developments raise fundamental questions about the economic consequences of AI progress. Unlike previous waves of technological change—mechanization, electrification, computerization—AI is distinctive in its potential scope. There exists no product or service that AI could not, in principle, create. This paper itself illustrates the point: it was written entirely by AI using engineered prompts, with code and documentation available at `https://github.com/chenandrewy/Prompts-to-Paper/`. The writing, mathematical derivations, and literature reviews were produced by large language models, demonstrating capabilities that would have seemed implausible just years ago.

The pace and breadth of AI advancement create uncertainty about labor market impacts. While technological progress has historically displaced some workers while creating new opportunities for others, AI's generality raises the possibility of more comprehensive disruption. Vinge (1993) introduced the concept of the technological singularity—a point where machine intelligence surpasses human capabilities across domains, potentially triggering an intelligence explosion with profound and unpredictable consequences. More recently, Bostrom (2014) and Russell (2019) have examined scenarios where advanced AI systems could fundamentally alter the economic position of human workers. Korinek and Suh (2024) analyze how output and wages respond when AI reaches human-level general intelligence, showing that wages could collapse under full automation unless counterbalanced by explosive growth.

Empirical evidence documents that automation already affects labor markets and asset prices. Acemoglu and Restrepo (2020) find that industrial robots reduce employment and wages in affected commuting zones. Zhang (2019) shows that firms with routine-task labor maintain a replacement option that hedges against unfavorable shocks, leading to lower expected returns. Knesl (2023) documents that firms with high shares of displaceable labor earn a return premium consistent with displacement risk. Babina et al. (2024) demonstrate

that AI investments affect firms' systematic risk profiles. These findings establish that technological change creates both winners and losers, with implications for how assets are priced.

This paper studies how AI-producing firms' stock prices reflect the possibility of transformative AI causing labor displacement. We develop a model where households face disaster risk: with some probability, AI capabilities advance dramatically, causing consumption to collapse as labor income falls while AI firms capture more economic surplus. Publicly traded AI equity serves as a partial hedge against this disaster. When AI transformation occurs, household consumption declines but AI dividends jump, creating negative covariance between the asset's payoff and marginal utility. This hedging value can justify high valuations even without beliefs about extraordinary future dividend growth.

The model generates several testable implications. First, AI equity should command a valuation premium that increases with perceived disaster probability, disaster severity, and household risk aversion. Second, this premium should be larger when AI ownership is more concentrated, as concentrated ownership amplifies market incompleteness and reduces households' ability to hedge through diversified portfolios. Third, the hedging channel predicts that AI valuations should be particularly sensitive to news about technological progress, labor market impacts, and regulatory frameworks affecting AI development. Fourth, cross-sectionally, firms with greater exposure to AI should exhibit lower expected returns when they provide better hedges against labor displacement risk.

We emphasize important caveats. We are not predicting that a negative singularity will occur. The probability and severity of AI-driven labor displacement remain deeply uncertain. Expert forecasts span wide ranges, and the path from current narrow AI systems to transformative general intelligence involves fundamental uncertainties about scaling laws, algorithmic breakthroughs, and societal adaptation. Rather, our contribution is to formalize a mechanism through which rational investors might value AI assets partly as insurance against technological disruption, even if they assign low probability to catastrophic scenarios.

We are also not claiming that this hedging value is already priced into current AI equity valuations. Observed prices reflect many factors beyond the mechanism we isolate: expectations about future earnings growth, competitive dynamics, regulatory uncertainty, and speculative demand. Empirically distinguishing the hedging channel from these other determinants presents substantial challenges. Asset prices are notoriously difficult to decompose into fundamental components, and the hedging effect we identify operates through subtle covariances that may be swamped by other valuation drivers. Our model illustrates a possible mechanism rather than providing a complete explanation for observed valuations.

The analysis connects to several strands of literature. The rare disaster framework builds

on Rietz (1988) and Barro (2006), who show that low-probability catastrophes can resolve asset pricing puzzles. Gabaix (2012) extends this framework to time-varying disaster severity, demonstrating how changing disaster risk generates volatile asset prices and return predictability. Gourio (2012) shows that disaster risk affects business cycles, with increases in perceived disaster probability leading to declines in output, employment, and asset prices. Our contribution lies in identifying a specific disaster scenario—transformative AI—where the disaster benefits some agents while devastating others, creating unusual hedging patterns distinct from traditional disaster models that assume symmetric harm.

The labor displacement literature documents how automation affects asset prices. Zhang (2019) examines how firms with automatable labor hedge against economic downturns, while we reverse the perspective: households may hold AI stocks to hedge against the very automation that threatens their labor income. Knesl (2023) focuses on how automation creates displacement risk that earns a return premium, whereas we examine how AI assets might provide insurance against aggregate technological displacement. Babina et al. (2024) study how AI investments affect individual firms' risk profiles; we examine how publicly traded AI assets serve as hedges for diversified households facing aggregate technological risk.

Our framework also relates to models of technological change and asset pricing. Santos and Veronesi (2006) show how the ratio of labor to financial income predicts stock returns when these income streams evolve stochastically. We extend this logic to a setting where technological shocks can dramatically and permanently shift the distribution of income between labor and capital. The model's emphasis on market incompleteness connects to work on portfolio choice with non-tradable labor income (Cocco, Gomes, and Maenhout, 2005; Benzoni, Collin-Dufresne, and Goldstein, 2007). However, where that literature typically assumes labor income risk is diversifiable or mean-reverting, we focus on aggregate, potentially permanent shifts in labor's economic position.

The analysis proceeds as follows. Section 2 develops the model, deriving equilibrium asset prices when households face disaster risk from transformative AI. Section 3 presents quantitative implications, showing how the hedging mechanism can generate high valuations under plausible parameters. Section 4 discusses the model's scope, limitations, and connections to broader questions about technological change and inequality. Section 5 concludes by examining implications for financial markets, policy design, and future research directions.

2 Model

We develop a deliberately stylized model that isolates the hedging mechanism at the core of our argument. The model abstracts from many realistic features of AI development

and financial markets, but this simplicity allows us to transparently demonstrate how AI assets can command high valuations even when their primary value derives from hedging catastrophic scenarios rather than expected cash flows.

The economy consists of two types of agents. AI owners hold concentrated positions in AI firms and are not marginal investors in the broader stock market. The representative household is the marginal investor whose consumption and preferences determine asset prices. This separation reflects the empirical reality that a small number of insiders and early investors hold substantial stakes in leading AI companies, while most households participate in AI equity through diversified portfolios. The representative household has constant relative risk aversion γ and time preference β. While the two-agent structure is a simplification—in reality, there exists a continuum of investors with varying exposures to AI—it captures the essential economic force: those who benefit most directly from AI advancement (through ownership of AI firms) differ from those who bear the primary consumption risk.

The validity of this separation rests on the observation that AI owners, by virtue of their concentrated holdings and often their direct involvement in AI development, have fundamentally different exposure to AI-related shocks than diversified households. As AI technology evolves, this separation may diminish if households accumulate larger AI equity positions or if labor markets adjust to preserve household income shares. However, at present, the concentration of AI ownership and the potential for rapid technological displacement justify treating these as distinct groups for analytical purposes.

Consumption growth for the representative household follows a simple process. In normal times, $\log \Delta c_{t+1} = 0$, so consumption remains constant. With probability p, a disaster occurs and $\log \Delta c_{t+1} = -b$ where $b > 0$. We interpret a disaster as a sudden, transformative improvement in AI capabilities—a localized singularity event. While the economy expands, the gains accrue predominantly to AI owners. For the household, labor income collapses as AI substitutes for human work across a broad range of tasks. The parameter b represents not merely a consumption decline but a comprehensive welfare loss that encompasses the destruction of meaningful work, social dislocation, and the psychological costs of obsolescence. In this sense, b is a sufficient statistic for the total utility impact of technological displacement.

The possibility of multiple disasters captures the ongoing uncertainty surrounding AI development. Even after one transformative breakthrough, further advances remain possible, each potentially eroding the household's economic position further. This specification connects to the broader literature on rare disasters, though our disasters have the unusual feature of benefiting some agents while devastating others. The repeated disaster structure also relates to compound option pricing: the value of hedging against the first disaster de-

pends on the distribution of subsequent disasters, creating path dependency in how investors value AI assets over time. Each disaster updates investor beliefs about both the probability and severity of future AI breakthroughs, generating time-varying risk premia even within our simple framework.

At time $t = 0$, no disasters have yet occurred, so $N_0 = 0$ where N_t denotes the cumulative number of disasters through time t. This assumption reflects our current position: transformative AI remains a future possibility rather than a realized event. The parameter p governs the arrival rate of these technological shocks, though we emphasize that the true probability is deeply uncertain and likely varies over time as AI research progresses.

The AI asset represents publicly traded AI stocks—firms producing semiconductors, cloud computing infrastructure, foundation models, and AI applications. Its dividend evolves according to

$$D_t = ae^{hN_t}C_t, \tag{1}$$

where C_t is aggregate consumption. The parameter $a > 0$ is small, reflecting that AI-related firms currently constitute a modest fraction of total economic output. While AI has captured substantial attention and investment, these firms' current dividends remain small relative to the broader economy. A reasonable estimate might place a in the range of 0.01 to 0.05, corresponding to AI firms representing perhaps one to five percent of aggregate dividends, though this varies depending on how broadly one defines AI exposure.

The exponential term e^{hN_t} captures how each disaster increases the AI asset's share of economic output. The parameter $h > 0$ reflects the elasticity of AI firm value with respect to technological breakthroughs. When a disaster occurs, AI capabilities advance dramatically, and firms controlling these technologies capture a larger share of total production. This can be understood through the lens of factor shares: as AI substitutes for labor, the capital share of output rises, and owners of AI capital appropriate more of the economic surplus. The parameter h thus encodes both the productivity gains from AI advancement and the degree to which these gains flow to AI firms rather than being competed away or captured by consumers through lower prices.

The multiplicative relationship between dividends and consumption, $D_t \propto C_t$, embeds an important assumption: AI firms' cash flows scale with aggregate economic activity. This seems reasonable for firms providing fundamental AI infrastructure—more economic activity generates more demand for computation, data, and AI services. However, the exponential term e^{hN_t} allows AI dividends to grow faster than consumption following disasters, reflecting the structural shift in how economic value is distributed.

We emphasize that this dividend process is a reduced-form representation of complex underlying dynamics. In reality, AI firms' valuations depend on expectations about future

technological progress, competitive dynamics, regulation, and myriad other factors. Our specification abstracts from these details to focus on the essential hedging mechanism: AI assets pay off precisely when household consumption collapses, creating negative covariance between the stochastic discount factor and AI returns. This negative covariance—the hallmark of a hedge—can justify high valuations even when expected dividend growth is modest.

The model deliberately omits several realistic features. We abstract from general equilibrium effects on interest rates, from the possibility that households might adjust by accumulating AI capital, and from the rich heterogeneity in how different households would experience an AI transition. We also treat the disaster probability p as constant, though in reality it likely varies with the state of AI research and may be fundamentally unknowable. These simplifications allow us to isolate the hedging mechanism while acknowledging that a complete treatment would require substantially more structure.

3 Results

We now derive the equilibrium price of the AI asset and demonstrate how the hedging mechanism generates valuations that can substantially exceed those implied by expected cash flows alone.

3.1 Asset Pricing Equilibrium

The representative household prices assets using the stochastic discount factor

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma}. \tag{2}$$

Given the consumption process, this discount factor takes two values: $M_{t+1} = \beta$ when no disaster occurs, and $M_{t+1} = \beta e^{\gamma b}$ when a disaster strikes and consumption falls by factor e^{-b}. The disaster state elevates marginal utility by factor $e^{\gamma b}$, reflecting the household's increased willingness to pay for consumption when it is scarce.

The risk-free rate satisfies $1 = \mathbb{E}_t[M_{t+1}]R^f$, yielding

$$R^f = \frac{1}{\beta[1 - p + pe^{\gamma b}]}. \tag{3}$$

Disaster risk depresses the safe rate because households engage in precautionary saving. When catastrophic consumption losses loom with probability p, the household accumulates safe assets, driving down their return. The magnitude of this effect depends on both the

disaster probability p and the severity b as amplified by risk aversion γ.

To price the AI asset, let $P_t^{(n)}$ denote its price at time t given that n disasters have occurred. The fundamental pricing equation is

$$P_t^{(n)} = \mathbb{E}_t[M_{t+1}(D_{t+1} + P_{t+1})]. \tag{4}$$

Conditioning on the current state $N_t = n$ and dividend $D_t = ae^{hn}C_t$, we expand this expectation over the two possible states next period. With probability $1 - p$, no disaster occurs, consumption and dividends remain constant, and the discount factor equals β. With probability p, a disaster strikes, consumption falls to $e^{-b}C_t$, dividends jump to $ae^{h(n+1)}e^{-b}C_t$, and the discount factor rises to $\beta e^{\gamma b}$. This yields

$$P_t^{(n)} = \beta(1-p)[D_t + P_{t+1}^{(n)}] + \beta p e^{\gamma b}[e^{h-b}D_t + P_{t+1}^{(n+1)}]. \tag{5}$$

The non-stationarity of the dividend process—through the unbounded counting variable N_t—might suggest that standard solution methods fail. However, the proportional structure of the problem admits a tractable solution. Define the price-dividend ratio $\phi^{(n)} \equiv P_t^{(n)}/D_t$. Dividing the pricing equation by D_t and rearranging yields

$$\phi^{(n)}[1 - \beta(1-p)] = \beta(1-p) + \beta p e^{(\gamma-1)b+h}[1 + \phi^{(n+1)}]. \tag{6}$$

We conjecture that $\phi^{(n)}$ is constant across all disaster states, that is, $\phi^{(n)} = \phi$ for all n. This ansatz proves correct because each disaster scales both current dividends and all future dividends by the same factor e^h, preserving the ratio between price and current dividend. Imposing $\phi^{(n+1)} = \phi$ and solving yields

$$\phi = \frac{\beta[1 - p + pe^{(\gamma-1)b+h}]}{1 - \beta[1 - p + pe^{(\gamma-1)b+h}]}. \tag{7}$$

This expression reveals the economic forces determining AI asset valuations. Define the effective discount factor

$$\tilde{\beta} \equiv \beta[1 - p + pe^{(\gamma-1)b+h}], \tag{8}$$

which combines three elements: pure time preference β, the probability-weighted no-disaster state $(1 - p)$, and the probability-weighted disaster state $pe^{(\gamma-1)b+h}$. The disaster term $e^{(\gamma-1)b+h}$ encodes the interaction between consumption risk and dividend growth. When disaster strikes, consumption falls by e^{-b}, raising marginal utility by $e^{\gamma b}$, while AI dividends jump by e^h. The net effect on the discount factor is $e^{\gamma b} \cdot e^h \cdot e^{-b} = e^{(\gamma-1)b+h}$.

The price-dividend ratio can be written as

$$\phi = \frac{\tilde{\beta}}{1 - \tilde{\beta}}, \tag{9}$$

making transparent that valuations explode as $\tilde{\beta}$ approaches unity. For convergence, we require $\tilde{\beta} < 1$, or equivalently,

$$h < b - (\gamma - 1)b + \frac{1}{\beta p} \ln \left(\frac{1 - \beta(1 - p)}{\beta p} \right). \tag{10}$$

When AI dividend growth in disasters (h) is sufficiently large relative to consumption decline (b) and risk aversion (γ), the present value calculation diverges because the asset's disaster payoffs compound faster than the household can discount them.

The hedging mechanism operates through the denominator $1 - \tilde{\beta}$. When $h < (\gamma - 1)b$, the AI asset hedges consumption risk: its dividends fall in disasters but not as severely as consumption, creating negative covariance between returns and the discount factor. This hedging value increases $\tilde{\beta}$ above β, compressing the denominator and elevating the price-dividend ratio. Even modest increases in $\tilde{\beta}$ generate large valuation effects because the ratio $\tilde{\beta}/(1 - \tilde{\beta})$ is highly convex near unity.

Consider the economic interpretation of the threshold $h = (\gamma - 1)b$. When h exceeds this value, AI dividends grow so rapidly in disasters that the asset amplifies rather than hedges consumption risk, commanding a positive risk premium. When h falls below this threshold, the asset provides insurance, and investors accept lower expected returns in exchange for disaster protection. The knife-edge case $h = (\gamma - 1)b$ produces a risk-neutral asset whose payoff scales exactly with the stochastic discount factor.

3.2 Quantitative Implications

To illustrate the model's predictions, we examine valuations under economically motivated parameters. We set $\gamma = 2$, reflecting moderate risk aversion standard in macrofinance, and $\beta = 0.96$, corresponding to an annual time preference of approximately four percent. The disaster probability $p = 0.01$ represents a one percent annual chance of transformative AI breakthrough, implying roughly ten percent probability within a decade and forty percent within fifty years—consistent with median expert forecasts for transformative AI timelines.

Table 1 presents price-dividend ratios across different combinations of consumption disaster severity b and AI dividend growth h. The table reveals several patterns. First, valuations rise sharply with both b and h as these parameters push the effective discount factor $\tilde{\beta}$ toward

unity. Second, a convergence boundary emerges around $b+h \approx 1.6$, beyond which valuations become unbounded. Third, even moderate parameter values generate price-dividend ratios that far exceed those of typical equity.

Table 1: Price-Dividend Ratios Under Alternative Scenarios

	AI Dividend Growth in Disasters (h)			
Consumption Disaster (b)	0.0	0.3	0.5	0.7
0.5	22.8	28.4	34.7	44.5
0.7	28.4	38.9	51.3	73.5
1.0	40.7	65.4	165.7	—
1.2	51.3	102.0	—	—

Note: Price-dividend ratios ϕ for $\gamma = 2$, $\beta = 0.96$, $p = 0.01$. Parameter b measures log consumption decline in disasters; h measures log AI dividend increase. Dashes indicate parameter combinations where $\tilde{\beta} \geq 1$ and valuations diverge. A consumption disaster of $b = 1.0$ corresponds to a 63% decline; $h = 0.5$ corresponds to a 65% dividend increase.

The baseline case with $b = 1.0$ and $h = 0.5$ generates a price-dividend ratio of 166. This represents a scenario where AI causes severe labor displacement—consumption falls by sixty-three percent—while AI firms capture moderately more of economic output, with dividends rising by sixty-five percent. The resulting valuation of 166 times current dividends far exceeds typical equity multiples, yet emerges purely from the hedging channel without requiring expectations of sustained dividend growth.

To understand this magnitude, consider the economic forces at work. The disaster probability of one percent per year creates meaningful consumption risk that households wish to hedge. When disaster strikes, consumption collapses while AI dividends jump, generating negative covariance between the asset's payoff and marginal utility. This negative covariance—the defining feature of a hedge—makes the asset valuable even though its expected dividend growth is modest. The effective discount factor $\tilde{\beta} = 0.994$ lies close to unity, compressing the denominator $1 - \tilde{\beta} = 0.006$ and producing the high valuation multiple.

The sensitivity to parameters reveals the model's economic content. Reducing the consumption disaster to $b = 0.7$ while maintaining $h = 0.5$ lowers the price-dividend ratio to 51. The smaller consumption decline reduces both the hedging benefit and the disaster-state discount factor, moderating valuations. Conversely, increasing AI dividend growth to $h = 0.7$ while keeping $b = 1.0$ pushes the model to its convergence boundary. The stronger AI payoff in disasters creates such powerful hedging value that the effective discount factor approaches unity.

The case of $h = 0$ merits particular attention. Here AI dividends track aggregate consumption without any disaster-specific jump. Yet the price-dividend ratio reaches 41 even

in this pure tracking scenario. This occurs because the asset maintains dividends when consumption collapses, providing insurance value despite zero expected dividend growth in disasters. The household values this insurance at forty-one times current dividends because the alternative—unhedged consumption risk—is sufficiently costly given the disaster probability and severity.

These quantitative results illuminate why AI equity might command high valuations in current markets. If investors perceive meaningful probability of transformative AI causing labor displacement, and if AI firms are positioned to capture more economic surplus in such scenarios, then the hedging mechanism alone can justify valuations that appear extreme by conventional metrics. The model does not require beliefs about explosive future earnings growth or irrational exuberance. Instead, rational hedging demand against a specific, identifiable risk can generate the observed valuation patterns.

The proximity to convergence boundaries carries implications for valuation stability. With $\tilde{\beta} = 0.994$ in the baseline case, the model sits just 0.6 percentage points from infinite valuations. Small changes in perceived disaster probability, expected consumption decline, or anticipated AI dividend growth can trigger large price movements. This suggests AI equity markets may exhibit substantial volatility as new information arrives about technological progress, regulatory frameworks, or labor market impacts. The convexity of the price-dividend ratio in $\tilde{\beta}$ amplifies these effects: parameter changes that would generate modest valuation responses in traditional assets produce outsized movements when the effective discount factor approaches unity.

The model also speaks to how valuations might evolve as AI development progresses. In early stages when transformative AI remains distant, low disaster probability p generates moderate valuations. As AI capabilities advance and transformation appears more imminent, rising p pushes valuations higher through the hedging channel. If and when a first disaster occurs, the model predicts that price-dividend ratios remain constant because the proportional hedging benefit persists. However, the absolute price level jumps by factor e^h as dividends increase, while aggregate consumption falls. This pattern—stable multiples but jumping price levels following technological breakthroughs—differs from traditional growth stock behavior where multiples compress as growth materializes.

The hedging interpretation also provides a framework for inferring market beliefs from observed valuations, analogous to extracting implied volatility from option prices. Given current AI equity valuations and estimates of γ, β, b, and h, one can back out the implied disaster probability p that rationalizes observed prices. If this implied probability appears inconsistent with expert forecasts or historical patterns of technological change, it suggests either that markets price factors beyond the model's scope or that valuations reflect mech-

anisms other than hedging demand. Such analysis could inform policy discussions about whether AI valuations represent rational risk management or speculative excess.

The model's simplicity—two agents, two states, one risky asset—allows transparent analysis of the hedging mechanism while abstracting from many realistic features. We have omitted general equilibrium effects on interest rates, heterogeneity in household exposures to AI risk, competitive dynamics among AI firms, and regulatory responses to technological displacement. Each of these extensions would enrich the analysis but potentially obscure the core insight: assets that pay off when consumption collapses can command high valuations through hedging demand alone, without requiring beliefs about extraordinary future growth. This mechanism, operating through the effective discount factor $\tilde{\beta}$, provides an alternative explanation for elevated AI equity prices that complements rather than contradicts growth-based narratives.

4 Discussion

We now interpret these results and discuss the model's scope and limitations.

4.1 Economic Interpretation

The hedging mechanism operates through the effective discount factor $\tilde{\beta} = \beta[1 - p + pe^{(\gamma-1)b+h}]$. When $h < (\gamma - 1)b$, the AI asset provides partial insurance against consumption disasters. Although AI dividends fall in disaster states—declining by factor e^{-b} while jumping by factor e^h, for a net change of e^{h-b}—they fall less severely than consumption. This negative covariance between returns and the stochastic discount factor creates hedging value that elevates the price-dividend ratio through the denominator $1 - \tilde{\beta}$.

The quantitative examples reveal that this mechanism can generate valuations far exceeding those of typical equity. With moderate parameters—consumption disasters of sixty-three percent, AI dividend jumps of sixty-five percent, and one percent annual disaster probability—the model produces price-dividend ratios around 166. This occurs without requiring beliefs about sustained dividend growth or irrational exuberance. Rational hedging demand alone, given meaningful probability of transformative AI causing labor displacement, can justify valuations that appear extreme by conventional metrics.

The proximity to convergence boundaries carries implications for market stability. When $\tilde{\beta}$ approaches unity, small changes in perceived disaster probability, expected consumption decline, or anticipated AI dividend growth trigger large price movements. The convexity of the price-dividend ratio in $\tilde{\beta}$ amplifies these effects. This suggests AI equity markets

may exhibit substantial volatility as new information arrives about technological progress, regulatory frameworks, or labor market impacts. The model thus predicts not only high valuations but also valuation instability.

The results connect to several strands of the asset pricing literature. The rare disaster framework builds on Rietz (1988) and Barro (2006), who show that low-probability catastrophes can resolve the equity premium puzzle. Our contribution lies in identifying a specific disaster scenario—transformative AI—where the disaster benefits some agents while devastating others, creating unusual hedging patterns. Traditional disaster models typically assume disasters harm all agents symmetrically, whereas we emphasize the distributional consequences of technological change.

The hedging interpretation also relates to long-run risk models following Bansal and Yaron (2004). In their framework, assets that covary negatively with consumption growth volatility command valuation premia. Our model can be viewed as an extreme version of this mechanism: the AI disaster represents a discrete jump in consumption volatility (from zero to catastrophic), and assets that hedge this jump become extraordinarily valuable. The difference is that we focus on tail risk rather than time-varying volatility of normal shocks.

4.2 Market Incompleteness and the Representative Household

The disaster magnitude b in our model represents the net effect on the representative household after accounting for both the AI transformation and any AI asset dividends the household receives. If markets were complete and the household could purchase shares in all AI assets—including private AI firms such as OpenAI, Anthropic, xAI, and DeepSeek—then the household could fully hedge and potentially benefit from the singularity. In that case, we would have $b < 0$, representing a sudden boom rather than a disaster.

The assumption $b > 0$ thus implicitly embeds market incompleteness. Many cutting-edge AI laboratories remain private, with ownership concentrated among founders, employees, and early investors. Most households cannot access these positions, leaving them exposed to labor displacement risk without commensurate capital gains. This incomplete markets structure is critical to our results: the hedging value of publicly traded AI assets arises precisely because households cannot fully insure against AI-driven consumption risk through private markets.

This interpretation connects to the broader literature on market incompleteness and technological change. Garleanu, Kogan, and Panageas (2012) develop a model where innovation creates displacement risk for incumbent capital owners, who cannot fully hedge because new technologies are not yet tradable. We apply similar logic to labor income: households face displacement risk from AI but cannot fully hedge because AI ownership is concentrated and

many AI assets are not publicly traded. The resulting hedging demand for available AI equity drives up valuations.

Historical episodes of technological change provide context for these mechanisms. The Industrial Revolution displaced artisans and cottage industry workers while enriching factory owners and investors in mechanized production. The rise of computing and automation in the late twentieth century eroded middle-skill employment while increasing returns to capital and high-skill labor. In each case, technological progress created winners and losers, with incomplete markets preventing full risk sharing. Our model formalizes this pattern in the context of AI, where the pace and scope of potential displacement may be unprecedented.

4.3 Model Limitations and Extensions

The model abstracts from many realistic features to isolate the hedging mechanism. We treat the representative household as a passive investor who cannot adjust labor supply, accumulate AI capital, or influence the pace of technological change. We assume AI owners are not marginal investors in the broader stock market, sidestepping general equilibrium effects on interest rates and asset prices. We take the disaster probability p as exogenous and constant, though in reality it likely varies with research progress and policy choices.

A more elaborate model would add detail to AI owners, private AI assets, and their interactions with the representative household. It could address questions such as: How does AI progress displace the representative household's wages? How do AI owners' incentives affect AI progress and market incompleteness? How do preferences and technology parameters affect the odds of a negative singularity? How does the nature and timing of technological progress affect equilibrium outcomes?

We have chosen not to pursue these extensions. While they would add realism, they would also decorate speculations with mathematics. The core economics—rare disaster risk, hedging motives, market incompleteness—would remain unchanged. The fundamental insight is that assets paying off when consumption collapses can command high valuations through hedging demand, and this insight does not require elaborate microfoundations for AI development or labor market dynamics.

Moreover, such extensions would substantially increase the cost of reading the paper. In our view, the benefit of reading should exceed the cost. A short model analysis allows room for the human-written appendix (Appendix A), which provides context on how this paper was generated using AI assistance. This transparency seems valuable given the nature of the project.

The model also abstracts from several economic mechanisms that could moderate or

amplify the hedging channel. We do not model competition among AI firms, which could dissipate rents and reduce h. We omit regulatory responses, which could redistribute AI gains or slow technological progress. We ignore the possibility that households might accumulate human capital complementary to AI, reducing effective b. We do not consider social insurance programs such as universal basic income, which could provide alternative hedges against labor displacement.

Each of these mechanisms merits serious analysis, but incorporating them would require taking strong stands on highly uncertain parameters and relationships. How rapidly will AI capabilities advance? How will labor markets adjust? What regulatory frameworks will emerge? What social insurance programs will be implemented? These questions involve deep uncertainty that cannot be resolved through theoretical modeling alone. Our approach is to isolate the hedging mechanism in a transparent framework, leaving these important but speculative extensions to future work.

4.4 Implications for Current Discourse

The model speaks to ongoing debates about AI valuations and economic impacts. Some observers argue that current AI equity prices reflect irrational exuberance or a speculative bubble disconnected from fundamentals. Others contend that high valuations are justified by expectations of transformative productivity gains and sustained earnings growth. Our analysis suggests a third interpretation: elevated AI valuations may reflect rational hedging demand against labor displacement risk, even without beliefs about extraordinary future growth.

This hedging interpretation has several implications. First, it suggests that AI equity prices contain information about market participants' beliefs regarding technological displacement. Just as option prices reveal implied volatility, AI valuations may reveal implied disaster probabilities. If observed valuations appear inconsistent with expert forecasts of AI timelines and impacts, it suggests either that markets price factors beyond our model's scope or that alternative mechanisms drive valuations.

Second, the hedging channel implies that AI equity serves a social insurance function. Households holding AI stocks partially hedge against labor income risk, reducing their exposure to technological displacement. This suggests that policies affecting AI ownership concentration—such as restrictions on private company investment, employee equity compensation rules, or public ownership proposals—have implications for household risk management beyond their direct effects on wealth distribution.

Third, the model's sensitivity to parameters near convergence boundaries suggests that

AI equity markets may exhibit substantial volatility. Small changes in beliefs about technological progress, regulatory responses, or labor market impacts can trigger large price movements. This volatility is not necessarily irrational or destabilizing; it may reflect rational updating about disaster probabilities in an environment of deep uncertainty. However, it does suggest that policymakers should consider financial stability implications of concentrated AI equity holdings and rapid valuation changes.

The analysis also connects to broader questions about technological change and inequality. If AI progress creates winners and losers, and if markets are incomplete such that losers cannot fully hedge, then the distributional consequences may be severe even if aggregate welfare increases. Our model formalizes this concern: the representative household faces disaster risk with magnitude $b > 0$ precisely because it cannot access the full portfolio of AI assets. Policies that broaden AI ownership or provide alternative hedges—through equity compensation, public investment vehicles, or social insurance—may reduce effective b and moderate both the hedging demand for AI equity and the welfare costs of technological displacement.

5 Conclusion

Our model demonstrates that AI assets may command valuations substantially exceeding those implied by expected dividend growth alone. This premium arises from hedging demand: when transformative AI threatens labor income, households value assets that pay off precisely in disaster states. The mechanism operates through the effective discount factor $\tilde{\beta} = \beta[1 - p + pe^{(\gamma-1)b+h}]$, which compresses the denominator of the price-dividend ratio when AI dividends covary negatively with consumption. With plausible parameters—moderate risk aversion, one percent annual disaster probability, and severe but not catastrophic consumption declines—the model generates price-dividend ratios around 166, far exceeding typical equity multiples.

This hedging interpretation differs fundamentally from standard technological adoption models. Traditional frameworks focus on productivity gains and expected earnings growth, treating technological change as a rising tide that lifts all boats. Our analysis emphasizes distributional consequences: AI progress creates winners and losers, with households facing labor displacement risk while AI owners capture economic surplus. The resulting hedging demand drives valuations through a channel distinct from growth expectations, suggesting that both mechanisms may contribute to observed AI equity prices.

The model also highlights how market incompleteness shapes these dynamics. The disaster magnitude b represents the net effect on the representative household after accounting for

any AI dividends received. If households could access the full portfolio of AI assets—including private firms such as OpenAI, Anthropic, xAI, and DeepSeek—they could potentially hedge completely and even benefit from technological transformation, making $b < 0$. The assumption $b > 0$ thus embeds incomplete markets: concentrated ownership and limited public trading of cutting-edge AI laboratories leave most households exposed to displacement risk without commensurate capital gains. This market structure is critical to our results. The hedging value of publicly traded AI equity arises precisely because households cannot fully insure through private markets.

This observation points toward financial market solutions to AI catastrophe risk that deserve more attention in policy discussions. The literature on AI existential risk, including influential work by Bengio et al. (2024), Bostrom (2014), Russell (2019), and Jones (2024), has focused primarily on technical AI safety, governance frameworks, and regulatory interventions. These approaches are essential, but they largely abstract from the role of financial markets in allocating and managing technological risk. Our analysis suggests that expanding household access to AI equity—through broader public offerings, employee ownership programs, or investment vehicles that track AI development—could provide partial insurance against labor displacement while reducing the concentration of AI-related wealth.

Such financial solutions offer an alternative to universal basic income and other transfer programs, though the two approaches are complementary rather than substitutes. Where UBI addresses displacement risk through ex-post redistribution funded by taxation, financial market solutions operate ex-ante through risk sharing embedded in asset ownership. The key economic distinction is that hedging through equity markets preserves incentives for efficient capital allocation while providing insurance, whereas pure transfer programs may create distortions if not carefully designed. However, the effectiveness of financial hedges remains limited by market incompleteness. If the most transformative AI capabilities remain concentrated in private firms, public equity markets cannot provide complete insurance regardless of how broadly shares are distributed.

The evolution of AI ownership structures will shape these dynamics. Currently, leading AI laboratories remain private or closely held, with ownership concentrated among founders, employees, and early investors. As these firms mature, they may pursue public offerings or alternative structures that broaden access. The transition from private to public markets could reduce effective b by allowing households to capture more AI gains, moderating both hedging demand and the welfare costs of technological displacement. However, this transition is not inevitable. Network effects, data advantages, and regulatory barriers may sustain concentrated ownership even as AI capabilities advance. Policy interventions that encourage broader ownership—such as preferential tax treatment for employee equity compensation or

public investment vehicles—could accelerate this transition and enhance welfare.

The model's implications for financial stability warrant consideration. With $\tilde{\beta} = 0.994$ in our baseline calibration, valuations sit just 0.6 percentage points from divergence. This proximity to the convergence boundary creates valuation instability: small changes in perceived disaster probability, expected consumption decline, or anticipated AI dividend growth trigger large price movements. Such volatility is not necessarily irrational—it may reflect rational updating about disaster probabilities in an environment of deep uncertainty—but it does suggest that concentrated AI equity holdings could amplify financial instability if valuations collapse rapidly.

The timing of potential AI disasters remains deeply uncertain. Expert forecasts span wide ranges, with median estimates suggesting roughly ten percent probability of transformative AI within a decade and forty percent within fifty years (Korinek 2023). Recent progress in reasoning models—including OpenAI's o1, DeepSeek's R1, and advances by Anthropic and Google—has compressed some timelines, but fundamental uncertainties persist about scaling laws, algorithmic breakthroughs, and the path from narrow capabilities to general intelligence. Our model takes no strong position on these timelines, treating the disaster probability p as a parameter that captures market beliefs rather than objective truth.

What the model does suggest is that even modest disaster probabilities can generate substantial hedging demand if the potential consumption impact is severe. A one percent annual probability may seem small in isolation, but it implies meaningful cumulative risk over investment horizons. Households facing such risk rationally seek hedges, and publicly traded AI equity serves this function to the extent that AI firms capture surplus during technological transformation. Whether this hedging demand reflects rational risk management or excessive pessimism about labor market adjustment cannot be determined from asset prices alone. However, the mechanism we identify—hedging against distributional consequences of technological change—operates independently of beliefs about aggregate productivity gains or existential risks to humanity.

The analysis points toward several directions for future research. First, incorporating general equilibrium effects on interest rates and wages would enrich the model's predictions about how AI development affects the broader economy. Second, modeling the endogenous evolution of AI ownership structures—including decisions by private firms about public offerings and by households about AI equity accumulation—could illuminate how market completeness changes over time. Third, examining how social insurance programs interact with financial market hedges would clarify the complementarity between these approaches to managing technological risk.

More fundamentally, the model suggests that asset pricing theory has a role to play in

debates about AI governance and economic policy. Financial markets aggregate information about technological risks and provide mechanisms for sharing those risks across agents with different exposures. Understanding how these mechanisms operate—and where they fail due to market incompleteness—can inform policy design. The hedging channel we identify is not a complete solution to AI-related displacement risk, but it represents an important dimension of how economies adapt to technological change. Recognizing this dimension may help policymakers design interventions that work with rather than against market forces, enhancing welfare while preserving the incentives for innovation that drive technological progress.

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A A Purely Human Perspective

The following is the README.md file from the GitHub repository:

```
# Prompts-to-Paper

Writes a paper about hedging a negative AI singularity, using AI.

- make-paper.py writes a paper

- plan0408-piecewise.yaml contains the prompts

- make-many-papers.py runs make-paper.py many times.

The README is entirely human-written.  Please forgive typos and errors.

-Andrew Chen, April 9, 2025

# Motivation

On March 8, 2025 I thought I should write a paper about hedging the AI
singularity.

I was worked up.  I had been repeatedly shocked by AI progress.  I was using
AI to prove theorems, vibe coding, and AI lit reviews in my daily life.  Six
months ago, I had thought each of these things is impossible.

What will happen in the next six years?!  Will my entire job be replaced by
AI? I have no idea.

But I do know that if there are huge AI disruptions, then tech stocks will
most likely benefit.  So if anything bad happens to my human capital, I
could at least partially hedge.  Strangely, I hadn't heard about this concept
before.

I asked a friend if he would be interested in working on this paper.
Unfortunately, he was busy.  So, I thought I should use AI to write the
paper.

## Inspiration

This project was inspired by Novy-Marx and Velikov (2025) and Chris Lu et
al.  (2024).  These projects use AI to generate massive amounts of academic
research.  My goal differs in quality over quantity.  I want to generate just
one paper, but one paper that (I hope) people find is worth reading.
```

This project was also inspired by Garleanu, Kogan, and Panageas's (2012) beautiful model of innovation and displacement risk. I read Garleanu et al. back when I was a PhD student and it has stuck with me.

Last, I drew from Hadfield-Menell and Hadfield (2018) and Bengio (2023), who apply ideas from economics to AI catastrophe risk. Hadfield-Menell and Hadfield (2018) explains the connection between incomplete contracting and AI alignment. Bengio (2023) frames AI catastrophe risk in terms of what I would call decision theory and human incentives---though the essay is written in plain English.

Previously, I dismissed "AI safety" as a politicized, activist cause. I still don't like the term "AI safety." "Safety" is such an excuse for exercising power over others.

But then the nature of AI changed. Things were progressing much faster than I expected. The Jan 15, 2025 episode of Machine Learning Street Talk with Yoshua Bengio left an impression on me. Bengio talked about AI catastrophe risk with no activism, no fear mongering. It was a straight, rational discussion of the seriousness of AI catastrophe risk.

The Paper Generation Process

Is this paper really written by AI?

I'd say the AI are junior co-authors.

If they were human, I would absolutely have to give o1, Sonnet, and ChatGPT Deep Research credit as co-authors. They did the math, writing, and literature reviews. Sonnet also wrote most of the code (via the Cursor AI IDE).

Of course, the prompts show I did substantial hand-holding. The Github commits show even more human labor. They tell the story of me getting to know my "co-authors." It was hard to communicate style issues and LaTeX instructions, leading to many, many commits.

To be honest, writing this paper would have been much easier if I had done more of the work myself.

But that can happen with human co-authors too.

Perhaps in the next few years, AI and humans will be equal co-authors.

Paper Iterations

Like human-written papers, the writing process was iterative. The first formalizations were terrible.

`plan0313-laborshare.yaml` (from March 13) contains prompts for a neoclassical growth model, where the capital share suddenly increases. ChatGPT-o1 patiently explained to me why this is a bad model.

me: > I thought there would be a wage risk effect that leads to higher investment for the more risk averse agent. High capital share means low or even no wage income.

ChatGPT-o1: > Below is an explanation of why one might **expect** a "wage-risk" channel in which **more** risk aversion could lead to **more** *additional* investment (relative to the no-jump benchmark) in the event that the capital share might jump to 1. However, this channel **does not operate** in the usual **representative-agent** version of the model---there, wage and capital income ultimately go to the *same* agent, so there is no meaningful "hedge" of wage risk.

I went through several iterations of the model with Claude 3.7 Sonnet (thinking mode) and ChatGPT-o1. The only derivations I did myself were to check o1's work, which I found to be quite reliable.

The final `plan0408-piecewise.yaml` uses a simplified Barro-Rietz disaster model, with two agents (though only one is relevant for stock prices). I slowly walk the AIs through the writing, using ten prompts, to maintain the writing quality.

Literature Reviews

A key step was generating lit reviews (`./lit-context/`) which were used as context in the prompts. I made lit reviews using ChatGPT's Deep Research (launched Feb 2025) until I ran out of credits. I used Claude Web Search (launched March 20, 2025) for the remainder.

These new products are a game changer. Both Novy-Marx and Velikov (2025) and Chris Lu et al. (2024) ran into hallucinated citations. Building off OpenAI Deep Research and Claude Web Search led to zero hallucinations in the paper drafts.

Still, I would occassionaly run into mis-citations. Every 15 to 20

citations, I would see a cite that mis- or overinterprets the cited paper. I suppose that's not so different than the human-written citation error rate. But I hate finding misinterpretations in the literature so I purposefully limited the number of cites in the paper.

AI Model Selection

I found it best to use o1 for theory, and Sonnet for writing. It's well-known that these are the strengths of these two models.

I briefly tried having Llama 3.1 405b do the writing. It was terrible! The cutting edge models seemed necessary.

I did not try many other models, in order to get this paper out quickly. Gemini 2.5's release, at the end of March 2025, was *hype*. I tried it out briefly and was impressed. But I gritted my teeth and ignored it. AI progress is too fast to update this paper continuously.

Picking the best of N papers

The writing quality varies across each run. Rather than try to prompt engineer an error free, insightful paper, I decided to just generate N papers and choose the best one.

5 drafts of the paper can be found in `./manyout0408-pdf/`. They're broadly similar and I would be willing to put my name on most of them.

I ended up choosing `paper-run-05.pdf`. The abstract overemphasizes the model, but otherwise I like the paper. I hope you find the paper worth reading.

Lessons about Research

A common response to Novy-Marx and Velikov (2025) is: "people are not ready for this." I heard concerns that peer review will be inundated with AI-generated slop.

Working on this paper gave me a different perspective. It made me think about the fundamentals. I think the fundamentals are the following:

1. Readers want to learn something interesting and true.

2. Readers don't want to check all the math.

3. A system of author reputations makes 1 and 2 possible.

AI-generated papers don't change any of these fundamentals. Critically, fundamental 3 made me quite wary of putting my name on AI slop. As a result, I don't think AI-generated papers will change much about peer review, at least not the current generation of AI.

Limitations of the Current AI (April 9, 2025)

This will likely be out of date by the time you read it.

But right now, AI is like a junior co-author with a talent for mathematics and elegant writing, but sub-par economics reasoning.

For example, Sonnet often fails to recognize that an economic model does not capture an important channel. This is common (no model can capture everything). The standard practice is to dance gingerly around the channel in the writing. A decent PhD student can recognize this. But Sonnet cannot.

AI also struggles to generate a satisfying economic model on its own. Its models are typically either too simplistic or too complicated.

I opted not to add empirical work or numerically-solved models. AI can do both, but both require connecting to the outside world, and a plethora of technical challenges.

Last, I often found the literature discussions insufficiently careful (as discussed above).

But how long will these limitations last?

The Future of AI and Economics Research (Speculative)

At some point, 2024-style economic analysis will be "on tap." You'll be able to go to a chatbot and ask "write me a paper about hedging AI disaster risk," and it will return you something like this paper (probably something much better).

"Economics on tap" could be a disaster for the economics labor market (could be). It certainly *will* be an extremely cheap substitute for at least some economists' labor. I suppose the questions is whether that will result in a strong substitution away from labor.

The optimistic argument is that AI also *complements* economists' labor. Perhaps, the number of economists will remain the same, but our research output increases in terms of both quantity and quality.

```
But there are reasons why total research output is limited.  Two key factors
in academic publishing are attention and reputation (Klamer and van Dalen
2001, J of Economic Methodology).  Readers can only pay attention to so
many scholars.  These scholars, in turn, can only pay attention to so may
projects.

I'm not saying that I *expect* a disaster for the economics labor market.  But
even if it's highly unlikely, it's still a scenario that economists should consider.
```

B Prompts Used to Generate This Paper

Each prompt consists of context and instructions. The context consists of the responses to the previous prompts, and may include literature reviews (all AI generated). For writing tasks (using Claude 3.7 Sonnet), a system prompt is also included.

For further details, see `https://github.com/chenandrewy/Prompts-to-Paper/`.

The system prompt and instructions are listed below.

System Prompt (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
You are an asset pricing theorist who publishes in the top journals
   (Journal of Finance, Journal of Financial Economics, Review of
   Financial Studies). You think carefully with mathematics and
   check your work, step by step.

Your team is writing a paper with the following main argument: the
   high valuations of AI stocks could be in part because they hedge
   against a negative AI singularity (an explosion of AI development
    that is devastating for the representative investor). This
   contrasts with the common view that AI valuations are high due to
    future earnings growth. Since the AI singularity is inherently
   unpredictable, the paper is more qualitative than quantitative.
   The goal is to just make this point elegantly.

Write in prose. Avoid bullet points and numbered lists. Use display
   math to highlight key assumptions. Cite papers using Author (Year
   ) format. Use we / our / us to refer to the writing team.

Be conversational yet rigorous. Favor plain english. Be direct and
   concise. Remove text that does not add value.
```

```
Be modest. Do not overclaim.

Output as a latex input (no document environment). Ensure bullet
   points are formatted in latex (\\begin\\{itemize\\} \\item "blah"
    \\item "blah" \\end\\{itemize\\}). Ensure numbered lists are
   formatted in latex (\\begin\\{enumerate\\} \\item "blah" \\item "
   blah" \\end\\{enumerate\\}). Ensure \\% is converted to \\\\%.
   But as a reminder, AVOID BULLET POINTS AND NUMBERED LISTS.
```

Instruction: 01-model-prose (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Draft the model description. Only describe the assumptions. No
   results or insights. Be modest.

Assume the reader is an asset pricing expert and knows standard
   results like the SDF and the 1 = E(MR).

Use the following outline:
  - The model is purposefully simple and captures the essence of the
      main argument
  - Two agents
    - AI owners
      - Fully invested in AI, not marginal investors in stock market
    - Representative household
      - Marginal investor in stocks: only their consumption matters
         for this analysis
      - CRRA = \\gamma, time preference = \\beta
  - Consumption growth
    - \\log \\Delta c\\{t+1\\} = 0 if no disaster
    - \\log \\Delta c\\{t+1\\} = -b if disaster (prob p)
    - A disaster is a sudden improvement in AI that is devastating
       for the household
      - Think of as a worst-case scenario for AI progress
      - Economy booms, but the value of AI is captured by the AI
         owners.
      - For household, labor is replaced by AI, so labor income
         plummets, as does consumption.
```

```
        - Also, way of life, meaning, is lost. Consumption fall can
           be thought of as a stand-in for these losses.
    - at t=0, no disasters have happened (singularity has not
       occurred)
      - Multiple disasters may happen, capturing ongoing uncertainty
         if a singularity occurs
  - AI asset
    - Captures publicly traded AI stocks
    - Dividend D\\_t = a e\\^\\{h N\\_t\\} C\\_t
    - Interpretation (discuss in prose)
      - a > 0 is small, AI stocks are currently a minor share of the
         economy
      - N\\_t is the number of disasters that have occurred up to
         and including time t
      - h > 0: each time a disaster occurs, the AI asset grows as a
         share of the economy
      - Intuitively, firms that provide semiconductors, data, AI
         models, etc. at least partially benefit from a sudden
         improvement in AI

Do not:
- Use bullet points or numbered lists
- Use any sectioning. No "\\section" or "\\subsection" commands.

Maintain academic rigor while emphasizing the underlying economic
   mechanisms. As you develop this model, clearly articulate the
   theoretical foundations in asset pricing theory, particularly how
    the stochastic discount factor operates in this context of
   technological disruption. Since the SDF can only be calculated in
    equilibrium, consider other measures as well (e.g. equivalent
   martingale). When describing the two-agent structure, be explicit
    about the economic implications of the separation between AI
   owners and households, especially regarding wealth distribution
   effects and market power dynamics. Justify why or why not this
   separation is valid and how it might evolve over time. For
   consumption growth, precisely define the economic interpretation
   of parameter 'b' in terms of welfare loss, not just as
   consumption reduction. Elaborate on the economic significance of
```

```
   repeated disasters, particularly how this relates to option
   pricing theory and path dependency in technological adaptation.
   Elaborate on the significance of repeated disasters on investor
   expectations and probability and distribution of the AI tail risk
   . When describing the AI asset dividend process, carefully
   articulate the economic meaning of parameter 'h' in terms of
   factor productivity and capital share of output. The relationship
    D\\_t = a e\\^\\{hN\\_t\\} C\\_t warrants deeper exploration of
   how technological progress alters factor shares in production.
   Ensure the model establishes clear mapping between theoretical
   parameters and potentially observable economic variables. Present
    this as a deliberate simplification that captures essential
   economic forces rather than a comprehensive representation.
   Discuss why a > 0 in depth, and provide an estimate ONLY IF YOU
   CAN REASONABLY do so of N\\_t.
```

Instruction: 02-result-notes (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Find the price/dividend ratio and risk premium of the AI asset at t
   = 0. The risk premium is the expected return (including dividends
   ) minus the risk-free rate. Derive the formulas, step by step,
   from first principles.

Do not:
- Restate the assumptions
- Assume any variable is constant or stationary (prove it)

When calculating expectations, explicitly account for the non-
   stationary nature of the dividend process and demonstrate why
   standard stationary assumptions fail in this context and what you
    should do instead. Apply the asset pricing kernel methodically,
   pay attention to convergence conditions for the infinite sum in
   the price calculation, ensure that your derivation highlights the
    uniqueness of outcomes related to disaster risk vs. expected
   growth, express all variables in closed form when possible. The
   mathematical elegance should be matched with economic intuition
   at each step.
```

Instruction: 03-table-notes (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Illustrate the results in ‘02-result-notes‘ with a couple numerical
   examples. Construct a quantitative illustration of the model’s
   implications through carefully chosen parameterizations that
   highlight economic significance. For the convergence analysis,
   select parameters that span theoretically meaningful thresholds.
   For example, consider gamma = 2, beta = 0.96, and p = 0.01. What
   values of b and h lead to convergence of the price/dividend ratio
   ?

When examining convergence conditions, interpret the economic
   meaning of parameter combinations that yield infinite valuations
   in terms of growth dominance versus discount rate effects. For
   each cell in your tables, provide precise economic interpretation
    of the magnitude, not just whether values are finite or infinite
   . Organize your sensitivity analysis to illuminate key
   comparative statics, particularly focusing on how disaster
   probability interacts with risk aversion to determine asset
   prices. This will provide insight into the quantitative
   significance of the hedging channel.
```

Instruction: 04-resultandtable-prose (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Convert the notes in ‘02-result-notes‘ and ‘03-table-notes‘ into
   prose. The prose is intended to follow ‘01-model-prose‘ and
   should flow naturally, ultimately to be in the same "Model"
   section.

The prose does not cover all results. It covers only the derivation
   and table for the price/dividend ratio. Synthesize the derivation
    and numerical results into cohesive scholarly prose that builds
   on the model description with mathematical precision and economic
    insight. The derivation should be easy to follow and self-
   contained. But do not output lecture notes. It should read like
   an academic paper. Fix notational issues like the re-use of the
   same variable name for different quantities. When presenting the
```

```
   price/dividend ratio formula, explicitly discuss the economic
   intuition behind each term, particularly highlighting how
   disaster risk creates a hedge value that potentially drives asset
    prices above standard present value calculations. Interpret the
   numerical findings in economic terms, explaining why certain
   parameter combinations yield explosive valuations and why. Draw
   explicit connections between the mathematical results and
   economic concepts such as risk aversion, time preference, and
   probability assessment of technological disruption, as well as
   how these parameters change over time as expectations evolve and
   AI continues to develop. When discussing the implications of the
   model, draw connections to historical episodes of technological
   valuation changes while maintaining theoretical discipline if you
    are reliably able to do so. Emphasize how the hedge value varies
    with model parameters in ways that might explain observed
   valuation patterns in technology sectors. Present the
   relationship between disaster probability and asset valuations in
    terms of implied disaster probability from observed market
   prices, similar to how options-implied volatility is extracted in
    derivatives markets.

This is the key text of the paper. Conclude the text by using the
   table to make the main argument. Avoid quantitative claims about
   the real world.

Style notes:
- The table should be clean and simple.
- Do not repeat information in `01-model-prose`.

Do not:
- Emphasize the infinite price/dividend ratio. That's not important.
- Use bullet points or numbered lists
```

Instruction: 05-discussion-prose (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Guidelines:
  - Market incompleteness is not explicitly modeled but important
  - Implicit in the disaster magnitude \\$b>0\\$
```

```
- 'b' is the *net* effect of (1) AI disaster and (2) AI asset
   dividend
- If markets were complete, representative household could buy
   shares in all AI assets (including private AI assets), and not
   only fully hedge but benefit from the singularity, implying \\
   $b < 0\\$ (a sudden boom, not a disaster)
- In reality, most households cannot buy shares in many cutting
   edge labs (e.g. OpenAI, Anthropic, xAI, DeepSeek), consistent
   with our model
- Relationship between results and historical episodes of
   technological change and capital deepening
- A more elaborate model would add detail to the AI owners,
   private AI assets, and their interactions with the
   representative household
- It could address questions like:
  - How does AI progress displace the representative household's
     wages?
  - How do AI owners' incentives affect AI progress and market
     incompleteness?
  - How do preferences and technology parameters affect the odds
     of a negative singularity?
  - How does the nature and timing of technological progress
     affect the model?
- But wouldn't this just decorate speculations with math?
  - The core economics (rare disaster risk, hedging motives,
     market incompleteness) will remain the same
- It would also be much more costly to read
  - In our view, the benefit of reading a paper should exceed the
     cost
- A short model analysis allows room for the human-written
   Appendix \\ref\\{app:readme\\}
Relate your findings with other insights like rare disaster events
   , equity premium puzzle, models of uncertainty, and related
   current economic discourse.
```

Instruction: 06-litreview-notes (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Find the most relevant papers and write a short two paragraph lit
   review based on the `*-prose` context. Explain how our work adds
   to the literature by proposing a new way to think about the
   valuation of AI stocks.

Be sure to cite:
  - Jones (2024) "AI Dilemma" and Korinek and Suh (2024) "Scenarios"
  - Babina et al (2023) "Artificial Intelligence and Firms'
     Systematic Risk"
  - Zhang (2019) "Labor-Technology"

Be careful to cite papers accurately. Make sure your claims match
   the `lit-*` context.

Do not:
  - Claim one paper builds on another unless you are sure this is
     true based on the `lit-*` context.
```

Instruction: 07-conclusion-prose (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Write a short "Conclusion" section.
- Review how the model demonstrates that AI assets may contain
   significant hedge value against technological disruption risk,
   creating a wedge between fundamental dividend-based valuations
   and market prices. Emphasize how this mechanism differs from
   standard technological adoption models by incorporating
   distributional effects and market incompleteness.
- End by discussing financial market solutions to AI catastrophe
   risk
  - These solutions are an alternative to UBI
    - Key economics: this hedge is limited by market incompleteness
- Will this change as AI companies become more public and
   transparent, if at all? What changes would enhance welfare and/or
    reduce uncertainty about technological outcomes?
  - These solutions are not discussed enough in the AI catastrophe
     risk literature (cite papers)
-  Discuss whether this is a long tail-risk or whether we could see
   an AI disaster in the near to medium term
```

```
  - Be very centrist
    - Avoid politically-charged topics (sovereign wealth funds,
       industrial policy, redistribution, extolling free markets)
  Be careful to cite papers accurately. Make sure your claims match
     the `lit-*` context.
```

Instruction: 08-introduction-prose (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Write the "Introduction" section, based on the `*-prose` context, as
    well as the `06-litreview-notes` that situates the paper within
   the economic literature while highlighting its distinctive
   contribution to understanding technological change and asset
   pricing.

Start with background. Describe how AI progress is happening quickly
    (release of reasoning models like ChatGPT-o1 and DeepSeek-R1,
   fast improvement in ARC-AGI Challenge, Waymo logging tens of
   millions of miles with self-driving cars), and investors may be
   concerned about their wages being displaced. Situate this within
   relevant economic context on labor market effects of productivity
    shocks while discussing the uniqueness of AI: technological
   change has occurred before, but AI is distinct because there is
   no product or service that AI could not, in principle, create. An
    example is the current paper, which is entirely written by AI,
   using a series of engineered prompts. Provide a link to the
   github site, which is https://github.com/chenandrewy/Prompts-to-
   Paper/. AI progress may also be incredibly sudden (the AI
   singularity).

Then describe what the paper does. It studies how AI stocks are
   priced, given that there is the risk that AI may destroy
   livelihoods and consumption. Note how the model generates
   testable implications about the relationship between perceived AI
    disruption risk and valuations of AI-producing firms.
Add caveats:
 We are not saying a negative singularity will happen
  But it is nevertheless important to consider this scenario
```

```
 We are also not saying that this hedging value is priced in already
    . Model illustrates a possible mechanism. Empirical limitations
    of estimating the hedging channel versus other determinants of
    asset prices.
Reference the broader literature on expectation formation in asset
   markets with fundamental uncertainty. End by incorporating the
   lit review (`06-litreview-notes`). When incorporating the
   literature review, position your work relative to both
   theoretical asset pricing models of rare disasters and empirical
   work on technology adoption and valuation cycles. Highlight how
   your approach provides new insights by explicitly modeling the
   hedge value of technological disruption.

Cite papers as appropriate. Ensure citations correspond to items
   from bibtex-all.bib. Make sure to cite Vinge (1993). Be careful
   to cite papers accurately. Make sure your claims match the `lit
   -*` context.

Do not:
- Discuss the role of human effort in making the paper
```

Instruction: 09-abstract-prose (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Write a less than 100 word abstract based on the `08-introduction-
   prose`, and `07-conclusion-prose`.

The abstract should:
- Make the main argument (AI valuations may be high because they
   hedge against a negative AI singularity)
- Define "negative AI singularity" after using the term (an
   explosion of AI development that is devastating for the
   representative investor)
- Touch on financial market solutions to AI catastrophe risk, in
   passing
- End with "Unlike previous work, this short paper is written by
   prompting LLMs."

Do not:
```

```
- Emphasize consumption
- Oversell or overinterpret the model
- Discuss the role of human effort in making the paper
```

Instruction: 10-full-paper (model: anthropic.claude-sonnet-4-5-20250929-v1:0)

```
Write a short paper titled "Hedging the AI Singularity" based on the
    `*-prose` context.

In page 1 of the introduction, include a footnote noting that "we"
   refers to one human author and multiple LLMs, and also that a
   purely human perspective is in \\hyperref[app:readme]\\{\\
   textcolor\\{blue\\}\\{Appendix \\ref\\{app:readme\\}\\}\\}.

Style Notes:
- Avoid bullet points and numbered lists
- No subsections (e.g. Section 1.2) though sections are OK (Section
   1)
- Don't say "in conclusion" or "in summary"

Output a complete latex document, including preamble. Use `template.
   tex` as a template. Keep the preamble, acknowledgements, and
   appendices as is.

Do not:
- Discuss the role of human effort in making the paper
```